i. **Porin**

Porin is a large transporter protein found in the outer membrane of plastids, mitochondria and bacteria which facilitates smaller molecules to pass through the membrane.

ii. **Aquaporin**

Aquaporin is a water channel protein embedded in the plasma membrane. It regulates the massive amount of water transport across the membrane (Figure 11.3). Plants contain a variety of aquaporins. Over 30 types of aquaporins are known from maize. Currently, they are also recognized to transport substrates like glycerol, urea, CO_2, NH_3, metalloids, and **Reactive Oxygen Species** (ROS) in addition to water. They increase the permeability of the membrane to water. They confer drought and salt stress tolerance.

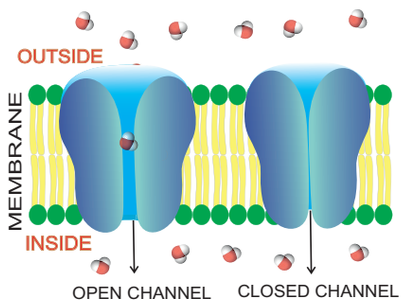


Figure 11.3: Aquaporin

II. Carrier Protein

Carrier protein acts as a vehicle to carry molecules from outside of the membrane to inside the cell and vice versa (Figure 11.4). Due to association with molecules to be transported, the structure of carrier protein gets modified until the dissociation of the molecules.

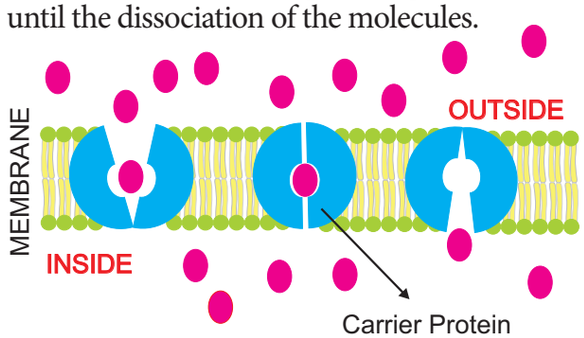


Figure 11.4: Carrier Protein

There are 3 types of carrier proteins classified on the basis of handling of molecules and direction of transport (Figure 11.5). They are, i) **Uniport** ii) **Symport** iii) **Antiport.**

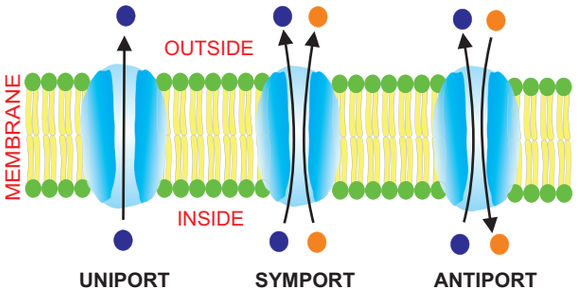


Figure 11.5: Direction of transport

i. ***Uniport***: In this molecule of a single type move across a membrane independent of other molecules in one direction.

ii. ***Symport or co-transport***: The term **symport** is used to denote an integral membrane protein that simultaneously transports two types of molecules across the membrane in the same direction.

iii. ***Antiport or Counter Transport***: An **antiport** is an integral membrane transport protein that simultaneously transports two different molecules, in opposite directions, across the membrane.



Discovery of Aquaporin



Peter Agre, discovered the "Water Pore" Aquaporin in RBC and received Nobel Prize for chemistry in 2003.

11.2.2 Active Transport

The main disadvantage of passive transport processes like diffusion is the lack of control over the transport of selective molecules. There is a possibility of harmful substances entering the cell by a concentration gradient in the diffusion process. But selective permeability of cell membrane has a great control over entry and exit of molecules. Active transport is the entry of molecules against a concentration gradient and an uphill process and it needs energy which comes from ATP. Passive transport uses kinetic energy of molecules moving down a gradient

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